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#include <stdio.h>
#include <string.h>

// Array to store input string
char a[16];

// Integer variable to store strlength of input string
int c;

// Use act array
char act[10];

// Use ac array
char ac[20];

// stk for stack
char stk[15];

void check();

int i = 0, j = 0, k = 0;

int main() {
    printf("GRAMMAR is \nE->E+E\nE->E*E\nE->(E)\nE->id\n");

    printf("Enter input string: ");
    scanf("%s", a);

    c = strlen(a);

    strcpy(act, "SHIFT->");

    printf("Stack \t Input \t Action\n");

    for (k = 0; j < c; k++, i++, j++) {
        if (a[j] == 'i' && a[j + 1] == 'd') {
            stk[i] = a[j];
            stk[i + 1] = a[j + 1];
            stk[i + 2] = '\0';
            a[j] = ' ';
            a[j + 1] = ' ';

            printf("\n%s\t%s\t%sid", stk, a, act);
            check();
        }

        else {
            stk[i] = a[j];
            stk[i + 1] = '\0';
            a[j] = ' ';
            printf("\n%s\t%s\t%ssymbols", stk, a, act);
            check();
        }
    }

    printf("\n");
}

void check() {
    strcpy(ac, "REDUCTION TO E");

    for (int z = 0; z < c; z++) {
        if (stk[z] == 'i' && stk[z + 1] == 'd') {
            stk[z] = 'E';
            stk[z + 1] = '\0';
        }
    }
}
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        printf("\n%s\t%s$\t%s", stk, a, ac);
        j++;
    }
}

for (int z = 0; z < c; z++) {
    if (stk[z] == 'E' && stk[z + 1] == '+' && stk[z + 2] == 'E') {
        stk[z] = 'E';
        stk[z + 1] = '\0';
        stk[z + 2] = '\0';

        printf("\n%s\t%s$\t%s", stk, a, ac);
        i = i - 2;
    }
}

for (int z = 0; z < c; z++) {
    if (stk[z] == 'E' && stk[z + 1] == '*' && stk[z + 2] == 'E') {
        stk[z] = 'E';
        stk[z + 1] = '\0';
        stk[z + 2] = '\0';

        printf("\n%s\t%s$\t%s", stk, a, ac);
        i = i - 2;
    }
}

for (int z = 0; z < c; z++) {
    if (stk[z] == '(' && stk[z + 1] == 'E' && stk[z + 2] == ')') {
        stk[z] = 'E';
        stk[z + 1] = '\0';
        stk[z + 2] = '\0';

        printf("\n%s\t%s$\t%s", stk, a, ac);
        i = i - 2;
    }
}
}
```